

GENERAL THETA FOUNDATIONS I: ACQUIRED GEOMETRY AND CAUSAL RESOURCE TRANSFER

QIAN QI

ABSTRACT. A causal observer must pay for knowing when to act as well as for remembering what its observations imply. We prove a joint acquisition–state law for experiments admitting a finite affine realization of their executable future tests. Under block contraction of the acquired-state kernel and uniform bounds on actual acquired densities, the optimal excess risk is of order $M^{-2/d}$ at a checkpoint and $(M - n)^{-2/d}$ for an autonomous observer making exactly n acquisitions. Constants are uniform in n ; all phases and controller states are charged. The lower follows from equivalence of report traces and actual-law small balls. The upper combines conditional-barycenter calibration, an exact continuation-value Jensen identity, a boundary-valid mass–curvature estimate, and an unequal integer allocation of states. The decision loss need not be quadratic. Gaussian hidden-state sensors and noncommuting quantum instruments verify the same theorem from their raw kernels. A stratified extension permits repeated transitions between pure-state atoms and full-dimensional laws. Numerical execution and causal simulation are separately certified, and a common scored experiment gives attained calibration and deficiency terms.

1. THE CAUSAL COMPLETION PROBLEM

A prepared experiment determines probabilities of executable future tests. It does not endow its observer with a real-valued posterior register, a free clock, or a chosen manifold. The foundational question is how actual acquisition and permissible causal processing determine the resolution of a finite observer and the risk of its decisions. We retain the order

$$\begin{aligned} \text{prepared kernels} &\longrightarrow \text{future tests} \longrightarrow \text{predictive quotient} \\ &\longrightarrow \text{acquired law} \longrightarrow \text{causal implementation and risk.} \end{aligned}$$

Statistical geometry is a derived object in this chain. A formally reachable set is not an acquisition distribution, and terminal quantization need not describe an executable online observer.

The central obstruction considered here has two parts. Information compressed at an intermediate cut must still support the next experiment, and the finite machine must know its position in the protocol. Conditional-barycenter pooling addresses the first part: each retained label is calibrated under the law of the implemented observer, rather than assigned an approximation to an inaccessible filter. Its loss is an exact Jensen defect of the optimal continuation value. The second part cannot in general be removed by reusing labels at different phases. When all report laws have a common positive support, an exact internal deadline forces distinct phase supports.

Theorem 2.2 joins these mechanisms. Its generality is a finite affine predictive class derived from positive experimental instruments; it is not limited to a probability simplex or a commuting measurement algebra. The hypotheses are uniform statements about each raw one-step acquired law and its stability, not properties of a selected optimal code. The decisive improvement is the

Date: 8 October 2026.

2020 Mathematics Subject Classification. Primary 62B05, 60G35; Secondary 93E20, 94A34, 81P45.

Key words and phrases. causal experiment, acquired geometry, posterior pooling, finite state, curvature, quantum instrument.

joint law

$$(1) \quad R_{\text{cp}}(n, M) - B_n^* \asymp M^{-2/d}, \quad R_{\text{aut}}(n, M) - B_n^* \asymp (M - n)^{-2/d}, \quad M \geq n + 1.$$

Here B_n^* is the optimal full-history risk in the same n -acquisition experiment. The constants are independent of n and M . An autonomous class with $M \leq n$ is empty under the exact-deadline convention. Thus horizon-uniform completion does not mean ignoring the clock: for example, uniform comparison at the same budget follows when $M \geq (1 + \eta)n$, with a constant depending on $\eta > 0$.

The proof assigns few labels to phases whose continuation values have little sensitivity and more labels to the final phases. Block contraction suffices; individual one-step coefficients may exceed one. A convex extension argument permits acquired laws to approach the boundary, which is essential for ordinary unbounded Gaussian reports. Two structurally different realizations follow: classical hidden-state sensing and noncommuting positive quantum instruments. Their predictive dimensions are obtained from executable tests and raw changes of variables. Neither realization is used to prove the general theorem. The singular extension treats actual mixtures of atoms and continuous laws, preserving their subprobability masses.

The posterior-pooling identity and its interior curvature argument are inherited from the preceding article, supplied intact as Supplement U [10]. The complete separated-refinement and continuation-cut results remain in Supplement T [11], and the block-filtering and stopped-task proofs remain in Supplement S [12]. The new claims here are the boundary-valid transfer, the sharp charged-clock law, its block-stable task-general proof, and the Gaussian and noncommuting-instrument verifications. The state-space language itself is classical [2, 1]; it is not claimed as new mathematics.

1.1. Prepared kernels and affine predictive realizations. All measurable spaces are standard Borel. Preparation and legal interventions specify positive normalized kernels $K_t^\lambda(dx', dy, dc \mid x, a)$. Failed observations, physical duration and intervention costs belong to c . Policies use only past reports, actions and the declared interface. They cannot read the unknown parameter λ . The rate theorems below fix a known prepared experiment; a Bayes statement for that preparation is not a learning theorem or a simultaneous sufficiency theorem for every parameter.

An executable test is a legal finite continuation with a bounded scored readout. Two histories at a phase are equivalent when they have the same legal interface and the same expectation for every such test. Appendix A gives measurable hypotheses for this quotient. The finite-dimensional hypothesis used in the main theorem is the following additional, verifiable closure property.

Definition 1.1 (Affine predictive realization). The experiment has a compact convex predictive carrier K of affine dimension $d \geq 1$, with a fixed affine coordinate identification $K \subset \mathbb{R}^d$. Every relevant future-test expectation is affine on K , and the tests separate its points. For every phase t and legal action a , the unnormalized report-and-continuation functionals depend affinely on the incoming state. Their normalization gives a jointly Borel report density $r_t(s, a, y)$ and updated state $F_t(s, a, y) \in K$, relative to a fixed report measure m^a . They realize the actual report and future-test probabilities. Mixtures of preparations have the corresponding barycentric state.

The conditions hold on all K , including mixtures created by compression, not only on the image of a preferred full-history policy. Write

$$(2) \quad J_t(s, a)(dy) = r_t(s, a, y)m^a(dy), \quad \Lambda_t(s, a) = \text{Law}_{Y \sim J_t(s, a)} F_t(s, a, Y).$$

One way to verify the definition is to exhibit a finite-dimensional span of executable tests, containing the constant test and invariant under the unnormalized report instruments, then evaluate a basis on histories and mixtures. Positivity and normalization descend from the original experiment. Neither covariance nor Fisher information is an input. Closure and the existence of

the jointly defined update are assumptions to verify, not consequences of finite rank alone. The actual acquired law at a cut is always the pushforward of the specified preparation and policy, never an arbitrary measure on K .

We fix a norm $\|\cdot\|$ on the translation space of K . Wasserstein distance W_1 uses this norm. Lebesgue density and strong concavity below use the displayed Euclidean coordinates; equivalence constants of these norms are fixed experimental constants. Affine coordinate changes alter the constants, not the asserted exponents or state counts.

1.2. Task and autonomous state model. There is a common nonempty finite action alphabet $\mathcal{A}$ at every phase. After exactly n physical calls, an irreversible terminal decision is scored. Its conditional expected loss $\ell(s, u) \in [0, H]$ is affine in s for each decision u ; a Borel minimizing decision exists. Put

$$(3) \quad g(s) = \min_u \ell(s, u), \quad V_n = g, \quad V_t(s) = \min_{a \in \mathcal{A}} \int V_{t+1}(z) \Lambda_t(s, a)(dz), \quad B_n^* = V_0(s_0).$$

The affine risks imply concavity. We assume g is L_g -Lipschitz and α -strongly concave in Euclidean coordinates, with $\alpha > 0$. The latter is needed only for the quantitative converse. Upper transfer and the exact Jensen identity remain valid without it. These assumptions cover more than quadratic losses, as shown after the main theorem.

An autonomous observer has at most M persistent labels. Its action, whether to stop, and terminal decision depend on the current label alone. The next label may use the current report and fresh independent randomization. At persistent cuts there is no readable elapsed time, old output tape, history, instruction pointer or persistent random seed outside this label. Temporary computation is erased. A fixed read-only programme may depend on the known kernels, preparation, horizon and budgets, but not on acquired data or an unknown parameter. Exact real instructions belong to the cardinality model only; Section 6 distinguishes finite-bit execution. The observer must itself stop after exactly n calls almost surely. A terminal request supplied by an external clock is a different resource model.

A checkpoint observer may use full history during acquisition and keeps only M terminal labels. Both classes optimize over all admitted full-history sensing policies, use the same preparation and scored task, and pay the same physical call cost. For a deterministic policy π , write $\nu_n^\pi = \text{Law}(S_n^\pi)$ and $B_n^\pi = \mathbb{E}g(S_n^\pi)$. Define the task distortion

$$(4) \quad D_M^g(\nu) = \inf_{\mathcal{P}: |\mathcal{P}| \leq M} \sum_{C \in \mathcal{P}} \left[\nu(C) g\left(\frac{\int_C s \nu(ds)}{\nu(C)}\right) - \int_C g(s) \nu(ds) \right].$$

Zero-mass cells are omitted. This is a distortion of the actual law. Independent seed conditioning gives

$$(5) \quad R_{\text{cp}}(n, M) = \inf_{\pi} \{B_n^\pi + D_M^g(\nu_n^\pi)\}.$$

For fixed terminal readouts, assigning a history to a least-loss readout is a Borel function of its predictive state: each conditional readout risk is affine in that state. Replacing any history-based randomized encoder by this assignment cannot increase risk, and then optimizing each cell readout gives the barycentric distortion above. Conditioning independent policy seeds completes the reduction; randomized decisions do not improve a linear conditional loss. This is a checkpoint relaxation, not purification of a clock-free autonomous machine. Policy-dependent summands in (5) are not minimized separately.

2. THE JOINT ACQUIRED-GEOMETRY TRANSFER THEOREM

The following assumptions are imposed before choosing the observer or a codebook. They use the actual one-step laws in (2).

Definition 2.1 (Clock neutrality and block-stable acquisition). There are constants $\rho < \infty$, $B < \infty$, an integer $b \geq 1$, $0 < \theta < 1$, and positive numbers $\beta_t \leq B$, such that:

(i) For all t, s, a , the report law $J_t(s, a)$ is equivalent to a fixed probability reference measure m^a . The same reference is used at every phase. Thus a report does not reveal a previously uncharged clock through disjoint supports.

(ii) For all t, s, a , $\Lambda_t(s, a)$ has density at most ρ relative to d -dimensional Lebesgue measure on K .

(iii) For all t, s, s', a ,

$$(6) \quad W_1(\Lambda_t(s, a), \Lambda_t(s', a)) \leq \beta_t \|s - s'\|, \quad \prod_{j=t}^{t+b-1} \beta_j \leq \theta$$

whenever the indicated block is present. The report kernels are jointly Borel and continuous in total variation in s , uniformly over the finite action family.

For a sequence of horizons, the same constants are required. The last incomplete block is controlled by B . The definition permits $\beta_t > 1$; it does not assume pathwise contraction of two filters on every report. The raw-kernel criteria in Section 4 verify the report continuity and Wasserstein property directly. A density bound is stronger than positive noise, but no compact interior carrier or lower bound on acquired density is required.

Theorem 2.2 (Charged-clock acquired-geometry transfer). *Consider a prepared experiment with Definition 1.1, the task and machine model of Section 1, and Definition 2.1. There exist $0 < c \leq C < \infty$, depending only on K , its norm, $d, \rho, B, b, \theta, L_g, \alpha$, such that for every $n \geq 1$ and every $M \geq 1$,*

$$(7) \quad cM^{-2/d} \leq R_{\text{cp}}(n, M) - B_n^* \leq CM^{-2/d}.$$

The exact- n autonomous class is nonempty precisely when $M \geq n + 1$. For every such M ,

$$(8) \quad c(M - n)^{-2/d} \leq R_{\text{aut}}(n, M) - B_n^* \leq C(M - n)^{-2/d}.$$

The constants may be decreased or enlarged once so that the same pair works in both bounds. The upper is realized by a deterministic observer with at most M total labels, including all controller phases. It uses no discarded history or uncharged exact state.

More generally, without the block-product restriction, put

$$(9) \quad a_t = L_g \prod_{j=t}^{n-1} \beta_j, \quad \gamma = \frac{d}{d+2}, \quad A_n = \sum_{t=1}^n a_t^\gamma.$$

The empty product is one. There is a constant C_0 depending only on K, d, ρ and the norm comparison such that

$$(10) \quad R_{\text{aut}}(n, M) - B_n^* \leq C_0 A_n^{1/\gamma} (M - n)^{-2/d}, \quad M \geq n + 1.$$

Thus the continuation-sensitivity profile, not equal phase allocation, determines the upper. Block stability makes A_n uniformly bounded.

The rate is attached to the dimension of an actually acquired law after a real report. The theorem does not replace an unavailable acquired law by the formal dimension of K : condition (ii) forces a full-dimensional probability distribution at each call. Section 5 replaces d by the largest actually acquired stratum dimension under explicit mass assumptions.

Corollary 2.3 (Resolution and nonquadratic tasks). *For target excess $\eta > 0$ below a fixed constant, the required checkpoint labels are of order $\eta^{-d/2}$ and the required autonomous labels are*

$$n + \Theta(\eta^{-d/2}).$$

Persistent bits are therefore $\lceil \log_2(n + \Theta(\eta^{-d/2})) \rceil$, not programme length or total bit complexity.

On a finite terminal audit alphabet, let Φ be a C^1 strongly convex function on its probability simplex and let

$$\ell(j, u) = \Phi(e_j) - \Phi(u) - \nabla \Phi(u) \cdot (e_j - u).$$

If the audit probability map $P : K \rightarrow \Delta$ is affine and injective, these bounded proper losses satisfy the task hypotheses, after harmless rescaling. For example $\Phi(u) = \sum_j u_j^2 + \zeta \sum_j u_j^4$, $\zeta > 0$, is nonquadratic. The theorem applies with constants depending on P and Φ .

Proof. The first assertion is the inversion of (7)–(8). Conditional expectation of the displayed loss is uniquely minimized by the audit distribution, and its Bayes risk is $\sum_j P_j(s) \Phi(e_j) - \Phi(P(s))$. Strong convexity and injectivity give strong concavity in s ; compactness and C^1 regularity give bounded loss and a Lipschitz Bayes risk. This is an example class, not a claim for every bounded loss. $\square$

2.1. An actual-mass and continuation-curvature bound. For a concave value v on K , a finite acquired measure λ and a partition $\mathcal{P}$, denote the nonnegative expression in (4) by $\mathcal{J}_v(\lambda, \mathcal{P})$. It is invariant under affine changes of predictive coordinates. Lemma 3.2 supplies a convex Lipschitz extension f of $-v$ to all of $\mathbb{R}^d$. Let $\kappa_f = \Delta f$ be its positive distributional curvature measure. For a grid of side h , with cell centers z_C , define

$$(11) \quad \mathfrak{C}_h(\lambda, f) = h^{2-d} \sum_C \lambda(C) \kappa_f(\overline{B}_{3\sqrt{d}h}(z_C)).$$

Only cells meeting the compact carrier occur. The estimate

$$\mathcal{J}_v(\lambda, \mathcal{P}_h) \leq C_d \mathfrak{C}_h(\lambda, f)$$

holds also for singular measures. It does not normalize their mass or smooth a switching set. A bound on acquired cell mass and a bound on continuation curvature are separate inputs. Under (ii), bounded overlap reduces this expression to $C\rho \text{Lip}(v)h^2$. Equation (11) is a useful sufficient local quantity, not a necessary-and-sufficient characterization or an assertion that every singular law has exponent $2/d$.

3. PROOF OF THE TRANSFER THEOREM

We first separate an exact causal identity from its analytic estimate. Throughout, conditional states are taken under the law generated by the observer being constructed.

Lemma 3.1 (All-incoming calibration and continuation identity). *Suppose the current labels have probabilities p_i and conditional predictive states q_i . At label i choose an action attaining the minimum in (3). Let W_t be the post-report state before pooling after call t , and let $\lambda_t = \text{Law}(W_t)$. Partition K into finitely many Borel cells $C_{t,j}$, and programme the next label j with*

$$q_{t,j} = \frac{\int_{C_{t,j}} z \lambda_t(dz)}{\lambda_t(C_{t,j})}.$$

An arbitrary state is assigned to an unused label. Then $q_{t,j}$ is the actual predictive state conditional on label j . If the terminal decision minimizes $\ell(q_{n,j}, u)$, its risk R satisfies

$$(12) \quad R - B_n^* = \sum_{t=1}^n \mathcal{J}_{V_t}(\lambda_t, \mathcal{P}_t).$$

Each law λ_t has density at most ρ under Definition 2.1(ii).

Proof. For any executable future test with affine expectation f , disintegration and the realization property give its expectation conditional on a retained label i , action a_i and report y as $f(F_{t-1}(q_i, a_i, y))$. Integration over every incoming edge assigned to cell $C_{t,j}$ yields $f(q_{t,j})$. Separation by tests proves calibration, and induction starts with $q_0 = s_0$. There is no need to recover a discarded history. The unpooled law is the mixture $\sum_i p_i \Lambda_{t-1}(q_i, a_i)$; hence it obeys the density bound. Bellman optimality gives

$$\mathbb{E}V_t(W_t) = \mathbb{E}V_{t-1}(Q_{t-1}), \quad Q_t = \mathbb{E}[W_t \mid Z_t].$$

Subtracting and summing proves (12). At the last cut the minimizing readout has risk $\mathbb{E}g(Q_n)$. The representatives are constants computed for this very machine, including all incoming labels, not for a different full-history policy. $\square$

Lemma 3.2 (Boundary-valid acquired-mass curvature estimate). *Let $K \subset \mathbb{R}^d$ be compact, convex, and have nonempty interior. If v is concave and Euclidean L -Lipschitz on K , then $-v$ has a globally convex L -Lipschitz extension f . For every finite measure λ on K and grid side $h > 0$,*

$$(13) \quad \mathcal{J}_v(\lambda, \mathcal{P}_h) \leq C_d h^{2-d} \sum_C \lambda(C) \kappa_f(\bar{B}_{3\sqrt{d}h}(z_C)).$$

If λ is a probability law with density at most ρ , then for every integer $L_0 \geq 1$ there is a Borel partition with at most L_0 cells such that

$$(14) \quad \mathcal{J}_v(\lambda, \mathcal{P}) \leq C_{K,d\rho} L L_0^{-2/d}.$$

Neither an interior support margin nor differentiability of v is required.

Proof. Put $f_0 = -v$. At each $x \in \text{int } K$ select all supporting slopes of f_0 . Their Euclidean norms are at most L : directional difference quotients can be taken in every direction at an interior point. Take the supremum, on all $\mathbb{R}^d$, of the resulting supporting affine functions. This supremum is finite, convex, L -Lipschitz and agrees with f_0 on $\text{int } K$; continuity gives equality on K . In particular, no assertion about arbitrary Lipschitz extensions preserving convexity is being used.

First suppose the extension is smooth. For a cell center z , subtract its tangent plane and call the resulting convex function w . Thus $w \geq 0$ and $w(z) = 0$. Put $R = 2\sqrt{d}h$. For any x in the cell, $B_{R/2}(x) \subset B_R(z)$. Writing av for the normalized spherical average, the submean inequality for w , radial monotonicity, and the divergence theorem give

$$w(x) \leq \frac{C_d}{R^d} \int_{B_R(z)} w \leq C_d \text{av}_{\partial B_R(z)} w \leq C_d R \text{av}_{\partial B_R(z)} \partial_r w = C_d R^{2-d} \int_{B_R(z)} \Delta f.$$

Here radial monotonicity gives $\int_{B_R} w \leq (R/d) \int_{\partial B_R} w$. The radial inequality is $w(z + R\omega) \leq R \partial_r w(z + R\omega)$. This argument also applies in dimension one, with the boundary integral interpreted as the sum over two endpoints. If b_C is the conditional barycenter, the affine tangent terms cancel and

$$\int_C f d\lambda - \lambda(C) f(b_C) = \int_C w d\lambda - \lambda(C) w(b_C) \leq \lambda(C) \sup_C w.$$

Mollify f on $\mathbb{R}^d$. The convex mollifications converge uniformly on bounded sets, and their positive Laplacian measures converge weakly to κ_f . The limsup of the mass of the radius- R ball is bounded by the mass of the closed ball of radius $3\sqrt{d}h$. Summing the finitely many relevant cells proves (13).

A Lipschitz convex function has uniformly bounded Laplacian mass on any fixed bounded region: choose a smooth nonnegative cutoff χ that equals one there and use

$$\int \chi d\kappa_f = - \int \nabla \chi \cdot \nabla f \leq L \int |\nabla \chi|.$$

For h bounded above, the enlarged cell balls have bounded overlap, and $\lambda(C) \leq \rho h^d$. Equation (13) therefore gives $C_{K,d} \rho L h^2$. Enclose K in a fixed cube and subdivide each coordinate into $\lfloor L_0^{1/d} \rfloor$ intervals. There are at most L_0 cells. Since $\lfloor L_0^{1/d} \rfloor \geq L_0^{1/d}/2$, one has $h \leq C_K L_0^{-1/d}$. This proves (14), including $L_0 = 1$, by changing the fixed constant. The extension is used only in this proof, not as an observer state. $\square$

Lemma 3.3 (Continuation sensitivity and integer allocation). *With a_t as in (9), $\text{Lip}(V_t) \leq a_t$ in the chosen norm. If $m \geq 1$ and $p = 2/d$, there are positive integers $L_1, \dots, L_n$ with $\sum_t L_t \leq n + m - 1$ and*

$$(15) \quad \sum_t a_t L_t^{-p} \leq 2^p \left(\sum_t a_t^{1/(1+p)} \right)^{1+p} m^{-p}.$$

Under (6), $\sum_t a_t^{1/(1+p)}$ is bounded independently of n .

Proof. The Wasserstein inequality and the minimum over a common finite action set show $\text{Lip}(V_t) \leq \beta_t \text{Lip}(V_{t+1})$; induction proves the first assertion. Concavity of V_t follows independently by taking the infimum of the affine risks of all fixed continuation decision trees. Lipschitz continuity and a finite minimum give a Borel minimizing action.

Put $w_t = a_t^{1/(1+p)}$, $A = \sum_t w_t$, and choose

$$L_t = 1 + \left\lfloor \frac{(m-1)w_t}{A} \right\rfloor.$$

When $m = 1$, $\sum_t a_t \leq A^{1+p}$. When $m \geq 2$, $L_t \geq (m-1)w_t/A$, so the left side of (15) is at most $A^{1+p}(m-1)^{-p} \leq 2^p A^{1+p} m^{-p}$. This counts one real label even at a very insensitive phase.

If $k = n - t$, decomposition into complete consecutive blocks gives

$$a_t \leq L_g \max(1, B^{b-1}) \theta^{\lfloor k/b \rfloor}.$$

Summation of the resulting geometric series proves the uniform bound. No single-report contraction is asserted. $\square$

Lemma 3.4 (Clock support converse). *Under Definition 2.1(i), an autonomous observer stopping after exactly n calls has pairwise disjoint positive-probability label supports at cuts $0, 1, \dots, n$. In particular, it has at least n nonterminal labels and at most $M - n$ possible terminal labels. The claim allows fresh independent transition randomization.*

Proof. Fix a label z appearing at two different cuts $s < t$. From that same label, construct a reference observer trace by drawing each next report from m^a , where a is the action selected by its current label, and by using its actual randomized label-transition kernel. For every finite number of future calls, the actual and reference trace measures are mutually absolutely continuous. Indeed, conditioning successively on each reference trace multiplies by strictly positive finite report densities almost everywhere. Mixtures over an unobserved incoming physical state or past history retain this equivalence. An unused value on a null report set is immaterial.

Starting at cut t and label z , the machine stops after $n - t$ calls almost surely. The same event has reference probability one, and hence actual probability one starting at cut s with label z . It would then stop before n , a contradiction. The argument also covers $t = n$, since stopping is a function of the label. Each of the $n + 1$ cuts has nonempty support, proving the counts. Conversely, a deterministic chain of $n + 1$ labels with any fixed legal sensing actions realizes the deadline. Its terminal forecast may be poor, but the class is nonempty. $\square$

Lemma 3.5 (Task curvature and actual-law converse). *For every full-history policy, including random policies, the actual terminal predictive law has density at most ρ . If its terminal decision uses at most L labels, then its risk is at least*

$$B_n^* + c_{d,\alpha} \rho^{-2/d} L^{-2/d}.$$

Proof. Condition immediately before the last action and report. For every selected incoming state and action the conditional post-report law has density at most ρ ; mixing preserves that bound. Full-history randomization can be included among the independent observed variables for this argument. Its optimal full-history risk is at least B_n^* by conditioning on independent seeds and minimizing over deterministic policies.

Let S be the resulting full-history terminal state, and $Q = \mathbb{E}[S \mid Z]$ for the terminal label. Affinity of the conditional loss implies that the best label readout has risk $\mathbb{E}g(Q)$. Strong concavity gives

$$\mathbb{E}g(Q) - \mathbb{E}g(S) \geq \frac{\alpha}{2} \mathbb{E}|S - Q|^2.$$

For any L representatives, their radius- r balls have total probability at most $\rho L v_d r^d$. Taking $r = (2\rho L v_d)^{-1/d}$ leaves probability at least $1/2$ outside their union, so the squared distortion is at least $r^2/2$. This applies also to randomized encoding, since Q still takes at most L values. No density lower bound or formal reachable-set packing is used. $\square$

Proof of Theorem 2.2. For the checkpoint upper, execute a Borel Bellman-optimal full-history policy and partition its actual terminal law using Lemma 3.2 with $v = g$. The readout at each conditional barycenter gives $CM^{-2/d}$. Lemma 3.5 gives the matching checkpoint lower.

For an autonomous upper, put $m = M - n$ and take the allocation in Lemma 3.3. Starting with the preparation, inductively construct the calibrated observer in Lemma 3.1, using at most L_t grid cells at phase t . Each grid is global on K and therefore defines the next-label rule for every possible incoming label and report. The conditional barycenters belong to K by convexity. Phase labels are disjoint, so the total state count is $1 + \sum_t L_t \leq M$. The actual density bound and the curvature estimate give

$$R - B_n^* \leq C \sum_t a_t L_t^{-2/d} \leq C A_n^{1/\gamma} m^{-2/d}.$$

Norm-equivalence constants are absorbed in C . This proves the weighted assertion and, under block stability, the horizon-uniform upper. There is no inaccessible shadow state in its execution.

Finally Lemma 3.4 restricts every autonomous observer to at most $M - n$ terminal labels. Lemma 3.5 proves the lower in (8). All optimization is over the declared task, preparation, action family and internal exact-deadline interface. $\square$

Remark 3.6 (Why the interface is part of the theorem). If the report at call t includes an exact time stamp t , or an external terminal request is provided for free, positive trace equivalence across phases fails. A machine can then reuse labels across phases and need not pay n distinct nonterminal states. Thus $M - n$ is not claimed for externally clocked filtering or for arbitrary stopping times. Likewise, without strong concavity a task may be insensitive to some acquired directions, and the stated full-dimensional lower need not hold. The mass-curvature upper remains meaningful in these cases.

4. TWO RAW-KERNEL REALIZATIONS

The same theorem applies to two different positive instrument structures. We first give a shared elementary stability estimate. This estimate is a sufficient coupling bound, not a sharp general contraction coefficient for filtering.

Lemma 4.1 (Instrument coupling). *Let states be either probability vectors with the ℓ^1 norm or finite-dimensional density matrices with the trace norm. Let C be a positive trace-preserving affine preparation channel with*

$$\|C(s) - C(s')\| \leq \kappa \|s - s'\|.$$

Let A_y be a positive linear instrument density, with $\int A_y m(dy)$ trace preserving. In the matrix case positivity on Hermitian inputs suffices for the argument; the realizations below are completely positive. Write $q = C(s)$, $u_y = A_y(q)$, $r_y = \text{tr } u_y$, and $F_y = u_y/r_y$. Then

$$(16) \quad W_1(\text{Law } F_Y, \text{Law } F'_{Y'}) \leq 3\kappa \|s - s'\|, \quad \text{TV}(\text{Law } Y, \text{Law } Y') \leq \frac{\kappa}{2} \|s - s'\|.$$

Here $\text{TV}(P, Q) = \sup_E |P(E) - Q(E)|$.

Proof. For a signed vector or Hermitian v , its positive and negative parts and positivity of the instrument give

$$\int \|A_y(v)\| m(dy) \leq \|v\|, \quad \int |\text{tr } A_y(v)| m(dy) \leq \|v\|.$$

The second inequality proves the report bound. Couple the reports by their common density $\min(r_y, r'_y)$ and couple the remaining masses arbitrarily. On the common part,

$$\min(r_y, r'_y) \|F_y - F'_y\| \leq \|u_y - u'_y\| + |r_y - r'_y|.$$

The state space has diameter at most two. The cost of the remaining mass is thus at most $\int |r_y - r'_y| m(dy)$. The total cost is at most $3\|q - q'\|$, which gives (16). Zero report masses may be assigned arbitrary normalized states on null sets. Strict positivity, when needed for the clock converse, is checked separately. $\square$

4.1. Controlled Gaussian hidden-state experiments. Fix hidden states $0, \dots, d$, a report dimension $m_{\text{obs}} \geq d$, and a nonempty finite family of sensor actions. A call first moves the hidden state through a known stochastic matrix T_t^a and then reports

$$(17) \quad Y_t \mid X_{t+1} = j, a \sim N(\mu_j^a, \Sigma_a), \quad Y_t \in \mathbb{R}^{m_{\text{obs}}}.$$

All reports, including large ones, are retained as physical acquisitions. There is no clipping, conditional acceptance or free resampling. The means for each action are affinely independent and Σ_a is positive definite. The parameters can range over a fixed compact family for which the singular values in the conditions below stay positive. Assume every entry of T_t^a is at least $\tau > 0$. Let

$$\kappa_t = \max_a \frac{1}{2} \max_{i,j} \sum_k |T_t^a(i, k) - T_t^a(j, k)|.$$

For predictive minimality assume that, at each phase, at least one of these legal matrices is invertible. This does not prescribe the action selected by the optimal observer.

Theorem 4.2 (Gaussian acquired-geometry law). *Suppose the sequence $\beta_t = 3\kappa_t$ obeys the block bounds of Definition 2.1. The experiment (17), with a terminal categorical audit and any loss in Corollary 2.3, verifies every hypothesis of Theorem 2.2 on $K = \Delta_d$. Consequently its optimized checkpoint and autonomous excess risks satisfy (7)–(8), uniformly in n, M . The acquisition policies may depend on every previously acquired report; no fixed sensor schedule is assumed.*

Proof. For an incoming belief p , put $q = pT_t^a$. The positive emission densities $f_j^a(y)$ give

$$r(q, y) = \sum_{j=0}^d q_j f_j^a(y), \quad u_j = F_j(p, a, y) = \frac{q_j f_j^a(y)}{r(q, y)}.$$

These formulas are obtained from the original transition and emission kernels. Unnormalized joint probabilities of any finite continuation are linear in p ; hence the affine realization and mixture identities hold on the entire simplex. Every Gaussian density is strictly positive, so all report laws are equivalent to, for example, one fixed nonsingular Gaussian reference measure on $\mathbb{R}^{m_{\text{obs}}}$ at all phases. A change from Lebesgue to that probability reference does not alter F or its acquired law.

Let $u_0 = 1 - \sum_{j=1}^d u_j$ and $z_j = \log(u_j/u_0)$. Define the full-row-rank $d \times m_{\text{obs}}$ matrix B_a with rows $(\mu_j^a - \mu_0^a)^T \Sigma_a^{-1}$. Then

$$(18) \quad z = B_a y + c_a + (\log(q_j/q_0))_{j=1}^d,$$

where $(c_a)_j = -\frac{1}{2}[(\mu_j^a)^T \Sigma_a^{-1} \mu_j^a - (\mu_0^a)^T \Sigma_a^{-1} \mu_0^a]$. Thus z has a mixture of $d+1$ nonsingular Gaussian densities, with covariance $B_a \Sigma_a B_a^T$. Their means stay in a fixed bounded set because $q_j \geq \tau$. This argument takes the Gaussian linear image in d log-likelihood coordinates; it does not require the report dimension m_{obs} to equal the acquired dimension. If h_q is this density, there are uniform positive constants such that

$$h_q(z) \leq C \exp(-c|z|^2 + C|z|).$$

The softmax map is a diffeomorphism from $\mathbb{R}^d$ to $\text{int } \Delta_d$, and its Jacobian is $\prod_{j=0}^d u_j$. Therefore the actual posterior density is

$$(19) \quad \frac{h_q((\log(u_j/u_0))_{j=1}^d)}{\prod_{j=0}^d u_j}.$$

Since $u_0 = (1 + \sum_j e^{z_j})^{-1}$,

$$(\prod_{j=0}^d u_j)^{-1} = (1 + \sum_j e^{z_j})^{d+1} e^{-\sum_j z_j} \leq C_d e^{C_d |z|}.$$

The Gaussian tail dominates this factor. Equation (19) is consequently bounded uniformly over all p , actions and phases, although its support approaches every simplex boundary. This is the reason for the global extension in Lemma 3.2. It is a bound on the actual law, not on a truncated or conditioned experiment.

The stochastic-matrix contraction in ℓ^1 is $\|pT - p'T\|_1 \leq \kappa_t \|p - p'\|_1$. To see this, decompose $p - p'$ into equal positive and negative masses; their normalized images are convex combinations of rows, whose diameter is twice the displayed row coefficient. Lemma 4.1 proves (6) and report continuity.

Finally the one-step Gaussian mixture law identifies q . Indeed, after dividing a putative linear relation between its component densities by their common Gaussian quadratic factor, one obtains a relation between exponentials with distinct linear exponents. Restrict to a line for which their slopes are distinct; successive limits along that line show that every coefficient vanishes. An invertible legal T_t^a then identifies p . Hence executable report tests separate the incoming beliefs. Equality of beliefs conversely fixes every continuation law. The terminal audit probability map is $p \mapsto p$, so the task hypotheses hold. These verifications are independent of the proof of Theorem 2.2. $\square$

The class includes uniformly positive, slowly mixing individual phases, provided sufficiently mixing phases recur in bounded blocks. For example in two states one can choose invertible symmetric transition matrices with row coefficients periodically 0.8, 0.8, 0.02. The largest one-step certified coefficient is $2.4 > 1$, but every complete three-step product is $0.3456 < 1$. This example illustrates block stability, not a claim that weak mixing alone implies the theorem. The strict analytic feedback advantage for the earlier affine sensor is retained in Supplement U; no strict Gaussian feedback advantage is inferred merely from the presence of multiple actions.

4.2. Noncommuting positive instruments. Let the physical Hilbert space have finite dimension $D \geq 2$. Choose a real orthonormal basis $H_1, \dots, H_d$ of traceless Hermitian matrices, $d = D^2 - 1$, and a number $s > 0$ such that $s \sum_i \|H_i\|_{\text{op}} < 1$. For $y \in [-1, 1]^d$, set

$$(20) \quad E_y = I + s \sum_i y_i H_i, \quad \mathcal{I}_{t,y}^a(\rho) = E_y^{1/2} \mathcal{C}_t^a(\rho) E_y^{1/2}, \quad m(dy) = 2^{-d} dy,$$

where

$$\mathcal{C}_t^a(\rho) = (1 - \alpha_t^a) \tau_t^a + \alpha_t^a U_a \rho U_a^*, \quad 0 < \alpha_- \leq \alpha_t^a \leq \alpha_+ < 1, \quad \tau_t^a \succeq \tau I.$$

The additive formula on trace-one states extends linearly by multiplying the first term by $\text{tr } \rho$. The channel is implemented by a classical mixture of preparation and a unitary action; each invocation, including a preparation outcome, is charged. The report instrument is positive and completely positive, and $\int E_y m(dy) = I$ makes it normalized. It defines positive classical causal report kernels $r(\rho, y) m(dy)$ and the deterministic state update $F(\rho, y)$; a density matrix is the verified predictive realization, not an extra observer register. Neither the matrices H_i nor the effects E_y commute in general.

Theorem 4.3 (Noncommuting-instrument realization). *Suppose $\beta_t = 3 \max_a \alpha_t^a$ satisfies the block bounds. Equip (20) with a finite informationally complete terminal audit and a loss from Corollary 2.3. Theorem 2.2 holds on the full density-matrix carrier with its trace norm and dimension $d = D^2 - 1$. No common eigenbasis is required, and all comparisons optimize over the declared history-dependent action policies.*

Proof. For $q = \mathcal{C}_t^a(\rho)$, let $r_y = \text{tr}(E_y q)$ and $\sigma_y = E_y^{1/2} q E_y^{1/2} / r_y$. The common strictly positive lower bounds on q and E_y give strictly positive report densities on the whole cube. The affine mixture and continuation-test identities follow directly from linearity of the unnormalized instruments. Known trace-preserving products represent every finite controlled continuation once its report-contingent policy is fixed.

We verify the acquired density without assuming it. For positive definite q, σ define

$$A(q, \sigma) = q^{-1/2} (q^{1/2} \sigma q^{1/2})^{1/2} q^{-1/2}.$$

It is the unique positive definite solution of $AqA = \sigma$, by squaring after conjugation with $q^{1/2}$. The inverse of $E \mapsto E^{1/2} q E^{1/2} / \text{tr}(Eq)$, on positive definite matrices with $\text{tr } E = D$, is

$$(21) \quad E = \frac{DA(q, \sigma)^2}{\text{tr } A(q, \sigma)^2}.$$

Indeed $E^{1/2} = \sqrt{D / \text{tr } A^2} A$, giving the prescribed normalized output; conversely uniqueness recovers any positive E . Both maps are smooth on their positive definite domains, and their derivatives on the trace hyperplanes are inverses. The affine map $y \mapsto E_y$ has full rank d . Since the permitted q and the closed report cube form a compact subset of these domains, the absolute Jacobian of $y \mapsto \sigma_y$ has a uniform strictly positive minimum. The bounded report density $r_y 2^{-d}$ and ordinary one-to-one change of variables yield a uniform acquired density bound in any fixed affine chart on density matrices. Cube boundaries have Lebesgue measure zero; no multiplicity or conditioning factor is suppressed.

The channel difference is $\alpha_t^a U_a(\rho - \rho') U_a^*$ and hence contracts the trace norm by α_t^a . Lemma 4.1 gives the required Wasserstein and report-TV estimates. The next-report density $\text{tr}(E_y q) = 1 + s \sum_i y_i \text{tr}(H_i q)$ identifies every traceless coordinate of q . The known affine channel is injective because $\alpha_t^a > 0$, so these executable tests separate ρ . Equality of density matrices fixes all future instrument probabilities, proving predictive minimality for this experiment.

For completeness, a finite separating audit is obtained from the effects

$$G_{i,\pm} = \frac{I \pm e H_i}{2d}, \quad 0 < e < (\max_i \|H_i\|_{\text{op}})^{-1}.$$

Their sum is I , and the probability differences are $e \operatorname{tr}(H_i \rho)/d$, an injective affine coordinate map. Corollary 2.3 gives bounded strongly concave task risks in all d directions. The audit is irreversible and separately charged in Section 6. The proof is finite dimensional throughout; it does not assert an unbounded-operator closure or a uniform-in- D constant. $\square$

5. SINGULAR ACQUISITION AND CHANGING RANK

A density on a full affine carrier is not needed for the exact causal identity. The following extension keeps track of the mass of each acquired component. Its constants may depend on the fixed number of strata and their geometry.

Theorem 5.1 (Finite stratified transfer). *Retain Definition 1.1, clock neutrality, the task hypotheses and the block Wasserstein bound. Replace the full-dimensional density condition by the following uniform statement for every incoming state and legal action. A type readable from the current report and incoming programmed state decomposes the actual post-report law into finitely many subprobability laws λ_e , supported on fixed compact convex affine carriers $K_e \subset K$. Carriers of dimension zero are singletons. A positive-dimensional carrier has dimension $r_e \leq r$, and λ_e has density at most ρ_e in its own affine coordinates. Immediately before the final action and report, uniformly over every incoming state and legal action, one fixed r -dimensional component has mass at least $w > 0$ and subdensity at most $\rho_* < \infty$. Here $r \geq 1$.*

Then the checkpoint and autonomous excess risks satisfy

$$R_{\text{cp}}(n, M) - B_n^* \asymp M^{-2/r}, \quad R_{\text{aut}}(n, M) - B_n^* \asymp (M - n)^{-2/r}, \quad M \geq n + 1,$$

with constants independent of n, M . An admissible lower constant is proportional to $\alpha w^{1+2/r} \rho_^{-2/r}$. All stored type identities are included in the label count.*

Proof. At a phase with a sufficiently large allocation L , reserve one label for each singleton and split the remaining labels among the finitely many positive-dimensional types, assigning a fixed fraction to each. Conditional barycenters are computed separately for each type and cell, pooling all its incoming edges. They remain in K_e by convexity and have the exact predictive meaning of Lemma 3.1. Intersecting a type with a grid cell produces a Borel next-label rule. Distinct types may have intersecting carriers, but the readable type is part of the rule, so no unobservable stratum oracle is needed.

The proof of Lemma 3.2 in dimension r_e works for a subprobability law without renormalizing it. Its contribution is at most $C_e \rho_e \operatorname{Lip}(V_t) L_e^{-2/r_e}$. Since $r_e \leq r$ and the number of types is fixed, the sum is at most $C \operatorname{Lip}(V_t) L^{-2/r}$. For the finitely many smaller allocations, use a single pooled label: the Jensen gap is at most $\operatorname{Lip}(V_t) \operatorname{diam}(K)$. Increasing the fixed constant yields the same $CL^{-2/r}$ bound for every integer $L \geq 1$. Thus there is not a mandatory number of type labels at every early phase. Lemma 3.3, with $p = 2/r$, gives the upper with exactly the same $M - n$ resource count.

For the lower, mixing over the actual history immediately before the last action preserves the chosen component's mass lower w and subdensity upper ρ_* . For any L representatives in the ambient space, their radius- u balls intersect that affine carrier in volume at most $L v_r u^r$. Choose $u = (w/(2\rho_* L v_r))^{1/r}$. At least $w/2$ of the unnormalized component remains outside, giving squared distortion at least

$$\frac{w}{2} \left(\frac{w}{2\rho_* L v_r} \right)^{2/r}.$$

Strong concavity supplies the factor $\alpha/2$. Use $L = M$ for checkpoint and Lemma 3.4 to use $L \leq M - n$ for autonomous risk. The result uses a genuine terminal acquired submeasure, not a normalized hypothetical stratum. $\square$

Corollary 5.2 (Validation mixed with noncommuting observation). *Modify the raw instrument in (20) as follows. Fix $0 < \vartheta < 1$ and a rank-one orthogonal resolution $P_1, \dots, P_D$ of I . After the charged channel $\mathcal{C}_t^a$, perform with probability ϑ the projective instrument $q \mapsto P_j q P_j$, reporting its type and j ; otherwise perform the continuous instrument (20). The same choice probability is used at every phase. Then Theorem 5.1 holds with $r = D^2 - 1$. Its terminal converse contains the factor $1 - \vartheta$ rather than discarding that mass by conditioning.*

Proof. The two instrument types have a common reference measure with positive weights on the continuous cube and every atomic report j . Uniform positivity of q gives positive probability to every such atomic report. Consequently the complete report laws remain equivalent at every phase, including following a pure-state report. The post-report state is exactly P_j on validation, and its other component has mass $1 - \vartheta$ and subdensity at most $(1 - \vartheta)\rho$. The ordinary instrument coupling in Lemma 4.1 applies to the combined instrument, with the same $3 \max_a \alpha_t^a$ bound. Repeated calls can thus move from an interior state to a pure-state atom and, after the next positive preparation channel, back to a full-dimensional continuously acquired state. The transitions are part of the raw experiment, not a static direct sum of unrelated tasks.

Theorem 5.1 applies with singleton carriers $\{P_j\}$ and the full density-matrix carrier for the continuous type. Substituting $w = 1 - \vartheta$ and $\rho_* = (1 - \vartheta)\rho$ gives

$$w^{1+2/r} \rho_*^{-2/r} = (1 - \vartheta) \rho^{-2/r}.$$

The type is read from the report; any type retained across a cut consumes an ordinary label. If ϑ were allowed to vanish at selected phases, its report support could encode a clock, so that alteration would require a new clock analysis. $\square$

The theorem allows boundary approach within each convex carrier, but not an arbitrary countable accumulation of strata. It also permits nonuniform one-step stability through the block condition. No dimension is assigned to a zero-mass reachable component. For a general singular law outside this finite class, (11) remains a proved upper criterion; the matching rate requires additional mass and curvature information.

6. EXECUTION RESOURCES AND COMMON-TASK TRANSFER

The cardinality theorem is not a total-bit-complexity theorem. We record the resources before combining errors. One call in the preceding sections means one actual instrument report, including failed or validation reports. A physical preparation or reset within that call is a separately recorded operation, not a free successful-sample oracle. The irreversible audit used to score the final forecast costs one additional measurement when physically implemented; it is not visible to the predictor before scoring. Thus n sensing calls correspond to $n + 1$ total measurements in such a realization. The difference is a fixed, explicit cost, not a hidden asymptotic convention.

6.1. Finite descriptions and numerical execution. For the construction of Theorem 2.2, the phase sets are part of the single persistent label set. Hence retained information costs $\lceil \log_2 M \rceil$ bits. At precision p , storing d coordinates, a finite action instruction and a terminal readout for each label costs at most $CM(dp + \log_2 |\mathcal{A}|)$ bits, in addition to the model description and integration programmes. A grid index can be computed in temporary workspace, which is erased at the next cut. The theorem alone does not bound the complexity of computing Bellman integrals or certified barycenters.

Proposition 6.1 (Certified numerical observer). *Consider a full-dimensional instance and a specified allocation L_t , with grid widths h_t . A programmed observer uses actual-law barycenters for its own programmed actions. Suppose these actions have Bellman slack at most γ_t uniformly over their labels. Suppose a numerical evaluation of the ideal post-report coordinates has sup*

error at most b_t except on an event of probability at most ξ_t under the ideal observer law. Let the conditional complete physical kernels, including the scored audit, admit a causal coupling with total mismatch probability at most $\sum_t \zeta_t + \zeta_{\text{audit}}$. This is a bound on physical kernels or complete scored laws, not merely separate report marginals. Let the terminal readout incur uniformly at most $\omega(u)$ additional conditional loss when its approximation is within u in the specified output norm. Then the implemented risk is at most

$$(22) \quad B_n^* + C \sum_t a_t L_t^{-2/d} + \sum_t \gamma_t + \omega(u) + H \left[\zeta_{\text{audit}} + \sum_t \left(\zeta_t + \xi_t + C_{K,d} \rho \frac{b_t}{h_t} \right) \right].$$

The bracket may be replaced by its minimum with one for the mismatch contribution. The digital machine has the same persistent labels. Additional programme bits, report precision, workspace and runtime must be supplied by the numerical certificates.

Proof. For the exact programmed actions, the proof of Lemma 3.1 has a nonnegative Bellman slack at each phase, adding at most $\sum_t \gamma_t$. Their actual all-incoming barycenters are recomputed using those actions; thus changing the action programme does not silently retain barycenters from a different law. The continuation Lipschitz and actual density estimates are uniform in the actions.

Couple the exact and digital executions. Before their first differing label or physical trajectory, they issue the same programmed action. Coordinate error can change a grid index only when an ideal coordinate lies within b_t of a grid hyperplane. In a fixed bounding cube there are at most $C_{K,d}/h_t$ such hyperplanes; the total volume of their slabs is at most $C_{K,d}b_t/h_t$. The unconditional ideal acquired law has density at most ρ , giving the stated probability. Intersecting this event with the absence of earlier disagreements can only decrease its unconditional probability; no density estimate conditional on survival is asserted. Add the exceptional numerical and physical-coupling events by a union bound. Off their union the terminal label agrees, and the loss increases by at most $\omega(u)$. On their union the bounded scored loss increases by at most H . This proves (22). $\square$

For a Brier audit the ideal label readout is the conditional mean, and changing it by an output vector e adds exactly $\|e\|_2^2$. More generally a C^2 generator Φ with bounded Hessian gives $\omega(u) \leq Cu^2$ for the proper losses above, by the integral remainder formula. For an arbitrary C^1 loss, a quadratic output-error bound is not automatic. This distinguishes task-norm output precision from coordinate precision used in cell selection. On strata the same proof is applied typewise with the corresponding intrinsic slab estimates, provided the numerical certificate also preserves the readable type.

6.2. Resource-certified causal morphisms.

Proposition 6.2 (Typed common-task composition). *Let a parameter-independent causal simulator implement every policy in a declared target class using the source experiment. Specify its strategy lift, report transformation, legal action interface, internal state, physical clock and task readout. Suppose its complete stopped and scored joint law is within total variation ε of the target law, uniformly over that class and the model parameter. Here $\text{TV}(P, Q) = \sup_E |P(E) - Q(E)|$. If the target predictor uses M labels and the simulator uses S labels, their direct causal implementation uses at most SM labels and transfers a common loss in $[0, H]$ with additive error $H\varepsilon$. All additional clock states must already be in S or be multiplied in separately. Acquisition, preparation and physical-time costs follow the simulator's declared cost map.*

For two composable certificates, complete-law errors add and state counts multiply. Calibration and numerical errors that are themselves complete-law mismatch bounds add to this error. Errors measured in a task norm must first be converted using the loss modulus; they are not interchangeable

with total variation. A reverse risk lower requires a reverse certificate covering every source policy being lower bounded.

Proof. Run the simulator and predictor in the product of their labels. The strategy lift defines the next legal source action from this product, and the report transformation supplies the target report before its prescribed deadline. The readout is part of the same scored law. For a bounded loss, the layer-cake representation bounds expectation difference by H TV. For composition, append the second causal simulator and use contraction of total variation by Markov kernels and the triangle inequality; multiply the three retained state components. Costs are accumulated according to the specified maps, not inferred from statistical distance. Applying the construction to every policy in a reverse certificate and then taking infima gives the stated condition for a lower. Without that quantifier, an upper simulator does not imply a converse. $\square$

This proposition is the elementary constructive comparison mechanism of classical experiment theory [1, 7], with the resource interface made explicit. It is not a new total-variation inequality. Neither a fixed-time certificate nor finite total-variation proximity proves an infinite-horizon, stopping-time or large-deviation equivalence.

6.3. An attained four-coordinate resource law. The following experiment shows that the geometric, calibration and deficiency coordinates can occur together in one scored task. It is a sharpness realization, not a claim that an arbitrary permitted error tolerance must incur a positive risk.

Take an engine from Theorem 2.2, with $n \geq 1$ sensing calls. Independently prepare a bit $b \in \{0, 1\}$ and an inaccessible calibration offset $v \in \{-\delta, \delta\}$ with equal probabilities. The known amplitude satisfies $0 \leq \delta \leq 1$. Before the engine, make one mandatory channel call: it reports b with probability $1 - 2\varepsilon$ and reports an erasure symbol with probability 2ε , where $0 \leq \varepsilon \leq 1/2$. Immediately after this report and before any engine observation, the observer must issue an irreversible estimate $\hat{b}$. This early scored output is write-only: it cannot later be read as an external register and has no feedback. It may use the current report in temporary workspace. The final forecast includes the engine decision and an estimate $\hat{v}$. The common loss is

$$(23) \quad (b - \hat{b})^2 + L_{\text{eng}}(\hat{u}) + (v - \hat{v})^2,$$

where $L_{\text{eng}}(\hat{u})$ denotes the actual engine audit loss, including its quantum audit outcome when appropriate. All three summands are scored under this one product preparation. The initial channel action is legal only at that initial interface; subsequent sensing actions are the engine actions.

Corollary 6.3 (Same-task acquisition–memory–error law). *Put $k = n + 1$, the number of calls before the irreversible terminal audit. There are constants independent of $n, M, \delta, \varepsilon$ such that*

$$(24) \quad \begin{aligned} \mathcal{R}_{\text{cp}}(k, M, \delta, \varepsilon) - B_n^* &\asymp M^{-2/d} + \delta^2 + \varepsilon, \\ \mathcal{R}_{\text{aut}}(k, M, \delta, \varepsilon) - B_n^* &\asymp (M - k)^{-2/d} + \delta^2 + \varepsilon, \quad M \geq k + 1. \end{aligned}$$

The source-to-exact-bit causal deficiency, with its mandatory early deadline, is exactly ε . If the final audit is counted as a measurement, total acquisitions are $N = k + 1$ and the autonomous geometric term is $(M - N + 1)^{-2/d}$ for $M \geq N$. No audit measurement or internal reset is omitted from the resource account.

Proof. On a revealed bit the early squared error can be zero, whereas on erasure its conditional variance is $1/4$. Thus the unavoidable bit risk is $\varepsilon/2$. The offset is independent of every available observation, so its unavoidable risk is exactly δ^2 . Giving the full-history engine policy all this independent side information cannot lower its engine risk below B_n^* : condition on that side

information and minimize over engine policies. The final acquired-density lower is uniform over all such policies.

For an autonomous procedure, the initial channel label cannot appear at an engine phase, since it would issue an illegal channel action. Within engine phases, the clock-support proof still applies. There are therefore at least k nonterminal labels and at most $M - k$ terminal labels. Lemma 3.5 proves the engine contribution, and the three lower terms add because they are components of the same loss under the same preparation. The analogous checkpoint argument uses M terminal labels.

For an upper, estimate a revealed bit by its value and an erased bit by $1/2$, issue that value irreversibly, and merge immediately into a single engine initial label. Estimate the offset by zero. No bit status is retained during engine sensing. Allocate $M - 1$ labels to the engine, including its initial label. Theorem 2.2 gives the upper because $(M - 1) - n = M - k$. The extra channel interface uses one label. This early-output interface is deliberately specified; the same count is not asserted for a task that requires the bit to be recalled at the terminal cut.

For the deficiency lower, regard b as an unknown parameter. The two source channel laws have total-variation distance $1 - 2\varepsilon$. The two exact target-bit laws have distance one. If a parameter-independent simulator had defect e for both parameters, contraction and the triangle inequality would give $1 \leq 2e + 1 - 2\varepsilon$, hence $e \geq \varepsilon$. Reporting a fair guessed bit on erasure attains $e = \varepsilon$ for each parameter. The guess is required before the engine or terminal audit, so subsequent observations cannot repair the deadline. Appending the same controlled engine cannot increase this complete-law defect. This proves the asserted attained deficiency and (24). $\square$

The offset coordinate represents an explicitly unobserved calibration ambiguity, not online estimation of an unknown kernel. The geometric term comes from actual sensing laws; the two independent coordinates establish additivity in one task. Their role does not convert arbitrary simulator-error allowances or numerical tolerances into information-theoretic lower bounds.

7. RELATION TO PRIOR RESULTS AND THE SCOPE OF THE THEOREM

The present result concerns a finite-state implementation of a complete experiment, rather than only an approximation to a Bellman function. The distinction does not make nearby belief-aggregation work irrelevant. Grover and Dimitrakakis [4], Theorem 1 and Lemma 1, use depth-dependent discretizations to control planning error and the size of a belief tree. Thus adapting resolution to continuation sensitivity is not by itself a new principle. Here the objects counted are deployed autonomous labels; actual conditional barycenters give a second-order task loss, and a clock converse matches that construction.

Li, Hammar and Bertsekas [8], Proposition 5, bound the error of a two-stage feature/belief aggregate model by an oscillation quantity divided by one minus the discount factor. Their subsequent lower-approximation condition is a statement about the aggregate cost function, not a decision-theoretic lower for every M -label observer. Our analysis uses the posterior law of the implemented observer itself and a lower from its acquired mass. The two constructions answer different questions; neither is described as a special case of the other without an explicit reduction.

Kara and Yüksel [5], Theorem 12, relate finite-window policy error to filter stability, with exponential convergence under stated conditions and a discount restriction. Demirci, Kara and Yüksel [3] obtain contraction-based average-cost optimality and approximation results. These are important precedents for using stability to suppress remote information. The finite exact-deadline task here is not an average-cost or infinite-horizon result: the new joint bound includes the necessary n clock states and the remaining $(M - n)$ resolution states, while an exact calibrated Jensen identity supplies the task error. The elementary factor-three instrument coupling is not claimed as a sharper replacement for those filter results.

Lyu, Suen and Zhang [9], Theorem 1 and Proposition 1, characterize static optimal interval partitions using conditional means and value curvature, including uniqueness under logconcavity. Consequently conditional-mean aggregation and curvature-sensitive partition design are established ideas. The local quantity (11) serves a different role: it bounds sequential losses under a machine-dependent law, permits nonsmooth continuation values and boundary approach, and is combined with explicit online state accounting. It is sufficient, not a characterization of all optimal partitions.

The operator/instrument language of Davies and Lewis [2], finite-dimensional Bayesian conditioning, supporting hyperplanes, small-ball quantization and Blackwell–Le Cam comparison [1, 7] are standard tools. The proof above supplies the specific statements used, rather than attributing those tools to the present work. The noncommuting inverse (21) and the Gaussian density calculation verify this theorem’s experimental hypotheses; they do not turn matrix parameter counting into an acquired-geometry theorem.

7.1. What is and is not combined. Theorem 2.2 combines a sharp clock converse with a horizon-uniform causal upper on a finite affine predictive class. Theorem 5.1 adds actual singular mass and the two raw-kernel theorems verify the same general hypotheses without reversing logical dependence. The exact identity is valid beyond the density and strong-concavity assumptions; the explicit exponent is not.

Supplement U retains the complete posterior-pooling article, including its strictly adaptive affine-sensor example and finite-precision arguments. Supplement T retains the complete acquired-refinement and serial-continuation article: its fixed-target adaptive acquisition and its noisy intermediate-cut obstructions are not replaced by the present fixed-deadline law. Supplement S retains the complete earlier block-stability, singular-transport and stopped-task proofs. These results have their original hypotheses and independent theorem numbers. Only the specific identity and analytic mechanisms cited in the introduction enter the present proof, and they are proved here in the form needed. Separate ordered-measurement, streaming-memory and finite-action research remains separate.

Several genuine generalization questions remain. Uniform acquisition density is a substantive smoothing condition; the mass–curvature expression suggests weaker hypotheses but does not by itself make them necessary. A general positive experiment need not have a finite affine realization or a Borel predictive quotient without additional measurability assumptions. The block condition is sufficient, not an intrinsic characterization of optimal causal completion, and it is not a theorem for unknown kernel learning. The exact- n task family permits a time-dependent terminal hidden state; it is not automatically a monotone fixed-target sampling law as n changes. Unbounded stopping and infinite time require new estimates. Finally, the exact finite-state construction and its conditional digital certificates do not determine the optimal joint region of programme length, workspace, numerical computation, reset budget and physical time. These are mathematical boundaries of the stated results, not substitutions of a different foundations problem.

APPENDIX A. PREDICTIVE QUOTIENTS AND MEASURABLE REALIZATION

For completeness we state a sufficient measurable construction before discussing its relation to the finite affine carrier. It does not assert that every measurable equivalence relation has a standard Borel quotient. The measurable ingredients below are standard; see also [6].

Fix a phase. Suppose its admitted history space $\mathcal{H}$ is a countable union of compact metrizable pieces. There is a countable determining family of bounded executable tests, together with countably encoded legal-interface coordinates. After rescaling, their evaluation map is

$$E : \mathcal{H} \longrightarrow [-1, 1]^{\mathbb{N}}.$$

Assume E is Borel on $\mathcal{H}$ and continuous on each compact piece, and that equality of these coordinates is exactly equality of all admitted future-test expectations and the legal interface. All raw report kernels and history concatenations are jointly Borel. Tests include first-report events in a countable generating algebra followed by every determining continuation test. This last closure specifies the conditional updating content of test equivalence.

Proposition A.1 (Realized minimal predictive quotient). *Under these hypotheses, $Q = E(\mathcal{H})$ is a standard Borel space, E realizes the predictive quotient, and E has a Borel section. Report kernels and predictive updates descend to jointly Borel kernels on Q ; update equality is asserted almost surely under the relevant report law. Every standard Borel sufficient state through which all the determining coordinates factor measurably has a unique measurable map to Q on its attained image. No operational information is added by the choice of section.*

Proof. The image of each compact piece is compact in the countable product. Their union Q is Borel in this Polish product and hence is standard Borel. Equality of coordinates is precisely the defining equivalence, not a finer relation involving unused tests.

Here is an explicit section construction. Select the least compact piece whose image contains q . In that piece the fiber $E^{-1}(q)$ is nonempty and compact. Take a countable dense sequence and successively choose the first closed ball of radius 2^{-k} meeting the fiber and all previously selected balls. Nonempty intersections are detected by membership in the image under E of a compact set, hence by a closed condition; at each step the countably many previous choices give a Borel partition of Q . The nested compact intersections have diameter at most 2^{1-k} and define one point of the fiber. Their centers converge to that point, so the selected point is Borel in q . This constructs a section $h(q)$.

Equality of first-report tests implies equality of report kernels at equivalent histories by the monotone class theorem. Equality of tests which first require a report in a generating event and then run a determining continuation test implies equality of its conditional continuation expectations almost everywhere under that common report law. The family is countable, so the entire evaluation of the updated history agrees almost everywhere. Define the quotient update by concatenating the report to $h(q)$ and evaluating E at the next phase. Its joint measurability follows from the section and concatenation. For any other representative it gives the same actual update almost surely. Integrating these per-history null sets against any admitted history distribution preserves that assertion; a universal pointwise version on every null report is not claimed. Arbitrary Borel assignments off legal or positive-probability interfaces complete the kernel.

If $T : \mathcal{H} \rightarrow S$ is another sufficient state with each coordinate of E a Borel function of T , the product of those coordinate functions gives a Borel map into $[-1, 1]^{\mathbb{N}}$ and takes $T(\mathcal{H})$ into Q . On that attained image it factors E and is unique because the coordinates separate points of Q . This is the asserted minimality. If the attained image of T is not known to be Borel, its trace sigma field is the stated domain; no additional Borel-image assertion is being inferred. $\square$

For a finite affine realization, evaluate a separating finite basis of the invariant test span. All other test coordinates are affine functionals of that evaluation. The positive unnormalized instruments give the explicit report and normalized update maps. The compact convex carrier K may contain mixture states or boundary limits not yet attained from one fixed preparation; it is a global legal extension on which the codebooks are executable. The actual laws ν_n^π and λ_t remain the corresponding preparation-and-policy pushforwards. The density or stratum hypotheses, rather than the mere size of K , supply the lower bounds.

In the Gaussian realization, histories of a fixed action word with reports restricted to closed bounded boxes form compact pieces, and the positive Bayes formulas are continuous there. Gaussian mixture identifiability gives separation. In the quantum realization the report cube,

finite validation alphabet and finite action words give compact pieces directly, and (21) and the separating report effects verify finite affine dimension. These explicit constructions avoid requiring a general quotient theorem for an arbitrary hidden-state or quantum process.

REFERENCES

- [1] D. Blackwell, *Equivalent comparisons of experiments*, Ann. Math. Statist. **24** (1953), 265–272, doi:10.1214/aoms/1177729032.
- [2] E. B. Davies and J. T. Lewis, *An operational approach to quantum probability*, Comm. Math. Phys. **17** (1970), 239–260, doi:10.1007/BF01647093.
- [3] Y. E. Demirci, A. D. Kara and S. Yüksel, *Average cost optimality of partially observed MDPs: contraction of nonlinear filters and existence of optimal solutions and approximations*, SIAM J. Control Optim. **62** (2024), 2859–2883, doi:10.1137/24M1643736.
- [4] D. Grover and C. Dimitrakakis, *Adaptive belief discretization for POMDP planning*, arXiv:2104.07276v1 (2021).
- [5] A. D. Kara and S. Yüksel, *Near optimality of finite memory feedback policies in partially observed Markov decision processes*, J. Mach. Learn. Res. **23** (2022), no. 11, 1–46.
- [6] A. S. Kechris, *Classical Descriptive Set Theory*, Graduate Texts in Mathematics 156, Springer, 1995.
- [7] L. Le Cam, *Asymptotic Methods in Statistical Decision Theory*, Springer Series in Statistics, Springer, 1986.
- [8] Y. Li, K. Hammar and D. Bertsekas, *Feature-based belief aggregation for partially observable Markov decision problems*, arXiv:2507.04646v1 (2025).
- [9] Q. Lyu, W. Suen and Y. Zhang, *Coarse information design*, arXiv:2305.18020v5 (2025).
- [10] Q. Qi, *General Theta Foundations I: Acquired Geometry and Causal Resource Transfer*, complete posterior-pooling article, integral Supplement U to the present manuscript, 2026.
- [11] Q. Qi, *General Theta Foundations I: Acquired Geometry and Causal Resource Transfer*, complete adaptive-refinement and continuation article, integral Supplement T, 2026.
- [12] Q. Qi, *General Theta Foundations I: Acquired Geometry and Causal Resource Transfer*, complete block-stability and singular-transport article, integral Supplement S, 2026.